

4. (a) Crude oil is a fossil fuel and is described as a non-renewable resource.

(i) Describe how crude oil was formed.

[2]

.....

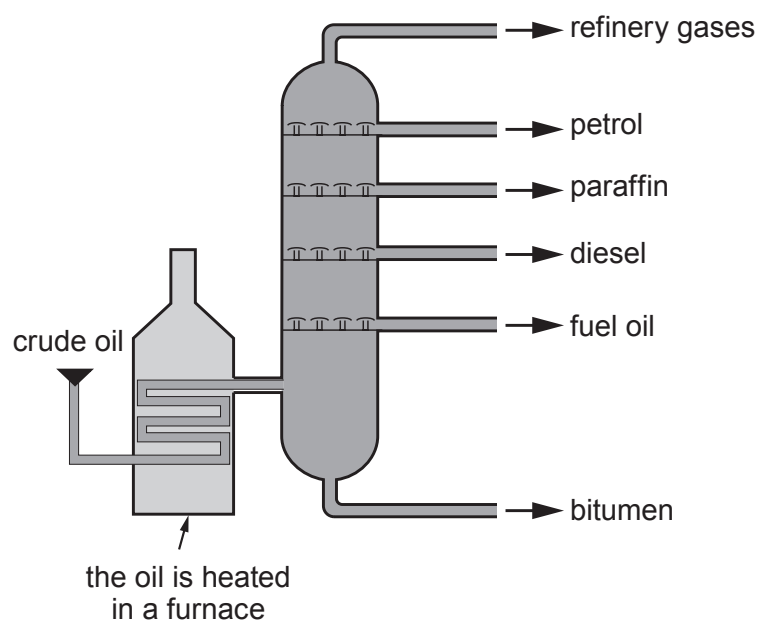
.....

(ii) Give the meaning of the term *non-renewable*.

[1]

.....

(b) To make crude oil more useful, it is separated into fractions.



(i) Complete the following sentences.

[2]

Crude oil is separated into different fractions by a process called fractional

.....

The fractions can be separated because they have different

.....

(ii) **Circle** the word which best describes crude oil.

[1]

mixture

element

compound



- (c) One of the fractions obtained from crude oil contains hexane, C_6H_{14} .

Calculate the percentage by mass of carbon in hexane.

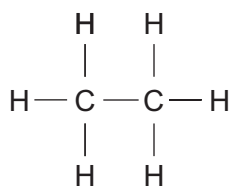
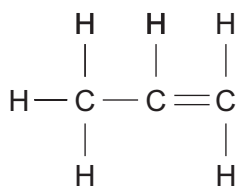
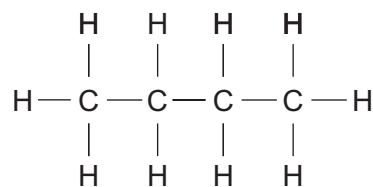
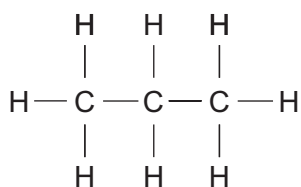
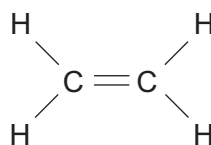
[2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %



(d) The diagram shows the structures of five different hydrocarbons, **A-E**.

**A****B****C****D****E**

Use letters **A-E** in your answers to parts (i) and (ii).

(i) Give the structure that represents propene. [1]

(ii) Identify the structures that fit the following descriptions. [2]

Hydrocarbons with the general formula $\text{C}_n\text{H}_{2n+2}$

Unsaturated molecules



- (e) Plastics are made from chemicals that are obtained from crude oil. Supermarkets in Wales were the first in the UK to charge their customers for plastic bags. This was to reduce the amount of plastic waste generated.

Give **two** methods of plastic waste disposal that lead to environmental problems. Explain the problem linked to each method. [2]

Method 1

Problem

.....

Method 2

Problem

.....

13

